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4.3.1 The SVD and the four fundamental spaces

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that that vector is equal to A times x . Okay? So we need to argue that we can also write that same vector as some, as the matrix U left times some vector, probably another vector. Okay? Now if you put x here as well, then you notice that all of this is some vector, and let's call it y . And therefore it seems like you're taking the linear combination of all of the columns of this matrix U right here. But if you look carefully

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Reading assignment

1/1 point (graded)

Read Unit 4.3.1 of the notes. [\[LINK\]](#)

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Homework 4.3.1.1

1/1 point (graded)
For Theorem 4.3.1.1, prove that $\mathcal{R}(A) = \mathcal{C}(A^H) = \mathcal{C}(V_L)$.

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Homework 4.3.1.3

1/1 point (graded)
Given $A \in \mathbb{C}^{m \times n}$, let $A = U_L \Sigma_{TL} V_L^H$ equal its Reduced SVD and $A = \left(\begin{array}{c|c} U_L & U_R \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H$ its SVD, and $r = \text{rank}(A)$. Then

- $r = \text{rank}(A) = \dim(\mathcal{C}(A)) = \dim(\mathcal{C}(U_L))$,
- $r = \dim(\mathcal{R}(A)) = \dim(\mathcal{C}(V_L))$,
- $n - r = \dim(\text{Null}(A)) = \dim(\mathcal{C}(V_R))$, and
- $m - r = \dim(\text{Null}(A^H)) = \dim(\mathcal{C}(U_R))$.

☒ done



Solution:

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

Homework 4.3.1.4

1/1 point (graded)
Given $A \in \mathbb{C}^{m \times n}$, let $A = U_L \Sigma_{TL} V_L^H$ equal its Reduced SVD and $A = \left(\begin{array}{c|c} U_L & U_R \end{array} \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(\begin{array}{c|c} V_L & V_R \end{array} \right)^H$ its SVD.

Any vector $x \in \mathbb{C}^n$ can be written as $x = x_r + x_n$ where $x_r \in \mathcal{C}(V_L)$ and $x_n \in \mathcal{C}(V_R)$.

TRUE

Answer: TRUE

Solution:

For a complete solution, see [the notes](#).

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